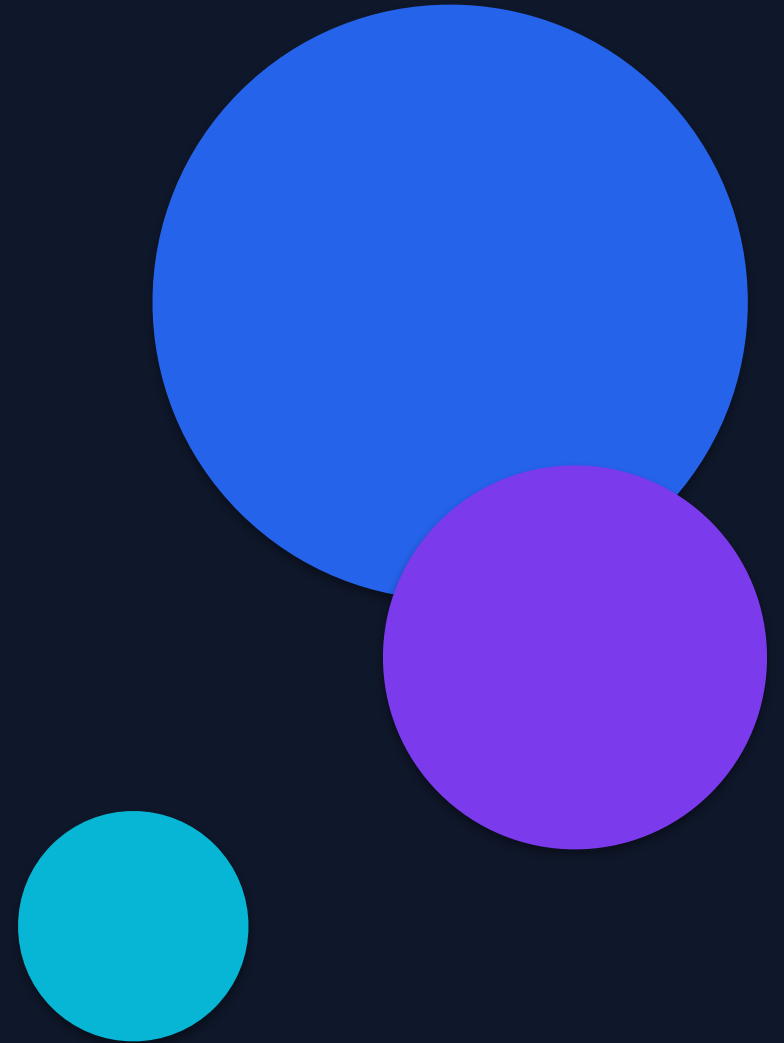


Prof. Bishal Sarkar

Lecture 02

Data to Intelligence

Generative AI & Agentic AI



From Data to Prediction

The basic pipeline behind a machine-learning problem.

Data

Features

Model

Prediction

ML Core idea

Transform real-world information into measurable inputs, learn from examples, and predict a new outcome.

Features and Labels

Two words every ML practitioner must understand.

X Features

The information used to make a prediction.

Examples:

- Area of a house
- Bedrooms
- Customer age
- Previous purchases

y Label / Target

The answer we want the model to predict.

Examples:

- House price
- Churn: Yes / No
- Spam: Yes / No
- Next month's sales

Features → what the model sees | Label → what the model predicts

What is a Model?

A learned relationship between inputs and an output.

Features (X)

Learned model

Prediction ($\hat{y}$)

f Think of it as a function

$$f(X) \rightarrow \hat{y}$$

Train Training

Learn useful parameters from examples.

Use Inference

Apply the learned model to new data.

Training vs Testing

A model must be evaluated on data it did not learn from.

TRAIN Training Set

Used to learn the model.

The model sees these examples during training.

TEST Test Set

Used to evaluate the final model.

The model should not use these examples to learn.

Training = learn | Testing = evaluate generalization

Classification

Predict a category or class.

2 Binary Classification

Two classes.

Spam / Not Spam
Fraud / Not Fraud
Churn / Stay

N Multiclass Classification

More than two classes.

Cat / Dog / Bird
Low / Medium / High

? Key question

“Which category does this new example belong to?”

Output = a class label

Regression

Predict a numerical value.

₹ House Price

₹72.5 lakh

₹ Sales Forecast

₹18.4 lakh

° Temperature

31.7 °C

Question: “What number should the model predict?”

Classification vs Regression

C Classification

Output: category

- Spam / Not Spam
- Disease / No Disease
- Customer segment

Think: WHICH ONE?

R Regression

Output: number

- Price
- Demand
- Temperature
- Revenue

Think: HOW MUCH?

Dataset Preparation

Real-world data rarely arrives ready for a model.

Raw data

Clean

Transform

Split

Ready for ML

1 Missing values

Find them and decide how to handle them.

2 Categorical data

Convert useful categories into model-friendly representations.

Train/Test Split + Preprocessing

The order matters.

Dataset

Train / Test Split

**Fit preprocessing on
Train**

**Transform Train +
Test**

✓ Key rule

Fit data-dependent preprocessing using training data, then apply it to the test data.

A Real-World ML Pipeline

From a business question to a prediction.

Business problem

Collect data

Prepare data

Train model

Evaluate

? Example

Predict whether a customer is likely to churn.

→ Output

Churn probability / class.

\$ Business action

Target the right customers with retention offers.

Practical — Load a Dataset using Python

We move from concepts to actual data.

01 Step 1 — Load

```
import pandas as pd  
  
df = pd.read_csv("dataset.csv")
```

02 Step 2 — Inspect

```
df.head()  
Df.tail()  
df.shape
```

03 Goal

Before modeling, understand what the dataset actually contains.

Practical — Basic Data Exploration

Ask questions before building a model.

01 Structure

```
df.shape  
df.info()  
df.head()
```

02 Statistics

```
df.describe()
```

Look at range, mean, spread and unusual values.

03 Data quality

```
df.isnull().sum()
```

Which columns have missing values?

First understand the data. Then decide what model to build.

Mini Challenge + Key Takeaways

You are the ML engineer.

CHALLENGE Scenario

An e-commerce company wants to predict whether a customer will purchase a product.

Discuss:

1. What could be the features?
2. What is the label?
3. Classification or regression?
4. What belongs in the training set?
5. What information should not be available at prediction time?

TAKEAWAYS Remember

- Features → inputs
- Label → target
- Training → learn
- Testing → evaluate
- Classification → category
- Regression → number
- Preparation → clean, transform, split